



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc CHEMISTRY

Programme	B. Sc. Chemistry				
Course Title	SCIENTIFIC COMMUNICATION, PUBLIC OUTREACH AND ENTREPRENEURIAL SKILLS				
Type of Course	SEC				
Semester	VI				
Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	2	-	1	45
Pre-requisites	1. Foundational knowledge in chemistry: Fundamental of chemical bonding and geometry of molecules, Concept of isomerism, Elements of symmetry of molecules 2. Proficiency in English to comprehend and engage in scientific writing and communication.				
Course Summary	This course equips participants with advanced technical writing skills, effective science communication strategies, and entrepreneurial insights, preparing them for diverse career pathways in the field of chemistry.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Recognize the importance of effective scientific communication in academia and in the society.	U	F	Instructor-created exams / Quiz
CO2	Acquire practical skills for the data presentation using 2D and 3D chemical structures and animations.	Ap	P	Practical Assignment / Observation of Practical Skills
CO3	Create scientific illustrations, diagrams, and video stories for science communication to the public audience.	C	M	Seminar Presentation
CO4	Comprehend the basics of entrepreneurship in science and identify opportunities in chemistry	Ap	C	Instructor-created exams / Assignments / Presentation
CO5	Apply entrepreneurial thinking in the development of research proposals	An	P	Writing assignments

* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)

- Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P)
Metacognitive Knowledge (M)

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	Science Communication to Public		8	15
	1	Importance of effective communication in academia and beyond and different forms of scientific communication	2	
	2	Tailoring Science for Public Audiences and adapting scientific language for non-experts,	2	
	3	Social media strategies for scientists, Science Journalism, Content Writing.	2	
	4	Interactive Public Presentations	1	
	5	Techniques for effective public speaking	1	
II	Technical skills for Academic Writing		9	18
	6	Structure and organization of scientific articles,	2	
	7	Paper formatting using MS Office, writing chemical equations and formulas, Understanding citation styles, reference management software	3	
	8	AI tools for literature review, content development, editing and data analysis.	2	
	9	Issues in scientific writing (plagiarism, authorship, ghostwriting, reproducible research).	2	
III	Entrepreneurial Skills for Scientists		9	18
	10	Introduction to Entrepreneurship in Science, identifying opportunities for entrepreneurship in chemistry	2	
	11	Basics of intellectual property rights, Understanding the patenting process	4	
	12	Overview of funding sources for entrepreneurial ventures, strategies for successful grant applications	3	
IV	Practical: Technical skills for Academic Writing		10	19
	13	Presentation of data in tables, figures and plots using excel / google sheet and using equations and functions in excel	1	
	14	Drawing 2D chemical structures using ChemSketch or Chemdraw	1	
	15	Creating 3D models of chemical structures for presentations and publications using Avogadro and JMOL; designing 3D molecular structures, measurement of bond length, bond angles and dihedral angles,	2	
	16	Visualizing atomic and molecular orbitals and analyse geometric and conformational isomers using JMOL	2	
	17	Exploring crystal structures, unit cell and symmetry operations in molecules using JMOL	1	
	18	Exploring the structure of protein and nucleic acid using JMOL	1	

	19	Creating scientific illustrations and diagrams (Inkscape, canva), video stories to communicate scientific information (Openshot, Kdenlive),	2	
V	Open Ended: Practical Application and Project Work		9	
	1	Case studies of successful scientific entrepreneurs	3	
	2	Public Communication Simulation: Role-playing scenarios for engaging with the public and media	2	
	3	Entrepreneurial Pitch Practice: Crafting and presenting a pitch for a scientific entrepreneurial idea	2	
	4	Writing and Presenting a Mock Research Proposal: Students develop a proposal integrating academic writing and entrepreneurial concepts	2	

Reference

1. Communication: A Practical Guide for Scientists, Laura Bowater, Kay Yeoman, Wiley-Blackwell, 2012
2. Robinson, M. S.; Stoller, F. L.; Costanza-Robinson, M. S.; Jones, J. K. Write Like a Chemist; Oxford University Press: New York, 2008.
3. Kovac, J. Write Like a Chemist: A Guide and Resource (Marin S. Robinson, Fredericka L. Stoller, Molly S. Costanza-Robinson, and James K. Jones). J. Chem. Educ. 2009, 86, 170.
4. Kelly, Kristine. "Translating Science: From Academia to Mass Media to the Public." In Taking Science to the People: A Communication Primer for Scientists and Engineers." University of Nebraska Press, 2010. ISBN: 9780803220522.
5. Avogadro software tutorial: <https://avogadro.cc/docs/>
6. Jmol Tutorial: https://wiki.jmol.org/index.php/Jmol_Tutorials
7. Communicating Science with social media: <https://medium.com/communicating-science-with-social-media>
8. Janet R. Morrow, Should You Become a Chemist Entrepreneur? Inorg. Chem. 2021, 60, 23, 17415–17418
9. Law Relating to Intellectual Property Rights, V K Ahuja, Lexis Nexis, 2017

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PS O5	PSO 6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1				2				3				3	3
CO 2				2				3				3	3
CO 3				1				3				2	3
CO 4				2				3				3	3
CO 5				2				3				3	3
CO 6				1				3				3	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Discussion / Seminar
- Midterm Exam
- Assignment (20%)
- Presentation (20%)
- Final Exam (40%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Presentation	End Semester Examinations
CO 1	✓			✓
CO 2		✓	✓	
CO 3		✓	✓	
CO 4	✓			✓
CO 5		✓	✓	

VALUE ADDED COURSES